বুলিয়ান উপপাদ্য

(i) $0+A = A$ → (i) $0 \cdot A = 0$

(ii) $1+A = 1$ → (ii) $1 \cdot A = A$

(iii) $A+A = A$ → (iii) $A \cdot A = A$

(iv) $A+\bar{A} = 1$ → (iv) $A \cdot \bar{A} = 0$

বিভাজন উপপাদ্য

(i) $A(B+C) = AB+AC$.

(ii) $A+BC = (A+B)(A+C)$

(iii) $\bar{A}+A\bar{B} = \bar{A}+\bar{B}$

(iv) $A \oplus B = A\bar{B}+\bar{A}B$.

সহায়ক উপপাদ্য

(i) $A(A+B) = A$.

(2) $A+AB = A$.

(iii) $A+\bar{A}B = A+B$.

(iv) $\bar{A}+AB = \bar{A}+B$.

(v) $A+\bar{A}\bar{B} = A+\bar{B}$

(vi) $\overline{\bar{A}} = A$.

OR operation:

(i) $0+0 = 0$. → সকল input 0 → output 0

(ii) $0+1 = 1$

(iii) $1+0 = 1$

(iv) $1+1 = 1$

And operation:

(i) $0 \cdot 0 = 0$

(ii) $0 \cdot 1 = 0$

(iii) $1 \cdot 0 = 0$

(iv) $1 \cdot 1 = 1$ সকল input 1 output — 1.

অতিরিক্ত বোর্ডন সূত্র

১। $A + AB = A$

২। $A(A+B) = A$

৩। $A(\bar{A}+B) = AB$

ডিমরগ্যানের উপপাদ্য

(১) $\overline{A+B} = \bar{A}.\bar{B}$	(১) $\overline{A+B+C} = \bar{A}.\bar{B}.\bar{C}$
(২) $\overline{A.B} = \bar{A}+\bar{B}$	(২) $\overline{A.B.C} = \bar{A}+\bar{B}+\bar{C}$

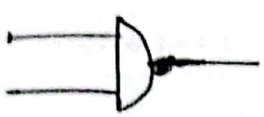
Logic gate

OR gate:



A	B	A+B
0	0	0
0	1	1
1	0	1
1	1	1

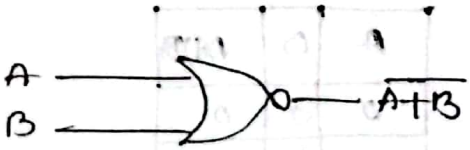
AND Gate:



A	B	AB
0	0	0
0	1	0
1	0	0
1	1	1

* NOR: NOT OR OR Gate এর আউটপুট কে উল্টাইলে যা আসবে:

A	B	$\overline{A+B}$
0	0	1
0	1	0
1	0	0
1	1	0



* NAND: NOT AND → AND Gate এর আউটপুটকে NOT করলে যা আসবে।

A	B	$\overline{AB}$
0	0	1
0	1	1
1	0	1
1	1	0



X-OR Gate:

Case- ❶ X-OR operation:

$0 \oplus X = X$

$0 \oplus 0 = 0$

$0 \oplus 1 = 1$

0 এর সাথে যার X-OR করবো Result— সেই আসবে।

$1 \oplus X = \bar{X}$

$1 \oplus 0 = 1$

$1 \oplus 1 = 0$

1 এর সাথে কারো X-OR করলে মান আসবে তার Prime.

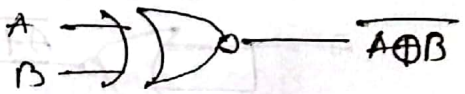


A	B	A⊕B
0	0	0
0	1	1
1	0	1
1	1	0

* X-NOR Gate:

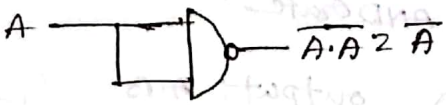
X-NOR operation : $\bar{A}B + A\bar{B}$

A	B	$\overline{A \oplus B}$
0	0	1
0	1	0
1	0	0
1	1	1

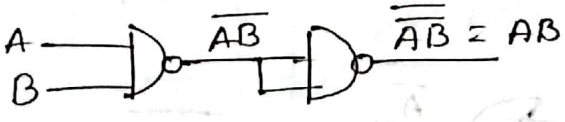


* Universal gate:

* NAND দিয়ে NOT :



* NAND দিয়ে AND: input: A, B. output: $AB = \overline{\overline{AB}}$



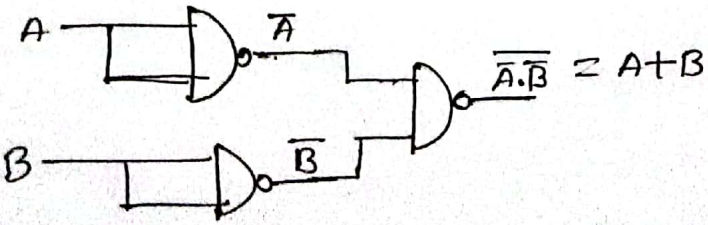
* NAND দিয়ে OR Gate:

$\overline{A+B} = \bar{A} \cdot \bar{B}$

input: A, B . output: $A+B$

$$\overline{\overline{A+B}}$$

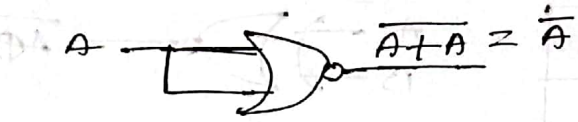
$$= \overline{\bar{A} \cdot \bar{B}}$$



* NOR Gate দিয়ে NOT Gate:

input: A, output: $\overline{A}$ [$A+A=A$]

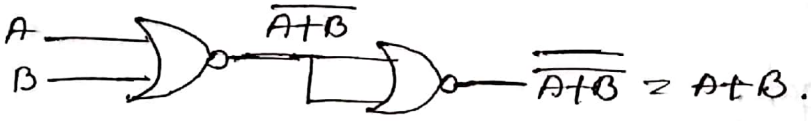
$= \overline{A+A}$



* NOR Gate দিয়ে OR Gate:

input: A, B output: $A+B$.

$= \overline{\overline{A+B}}$

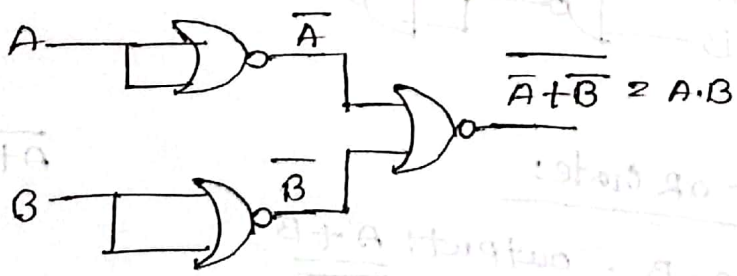


* NOR Gate দিয়ে AND Gate:

input: A, B output: $A \cdot B$

$= \overline{\overline{A \cdot B}}$

$= \overline{\overline{A}+\overline{B}}$



$0 \rightarrow \bar{A}, \bar{B}$
$1 - A+B$

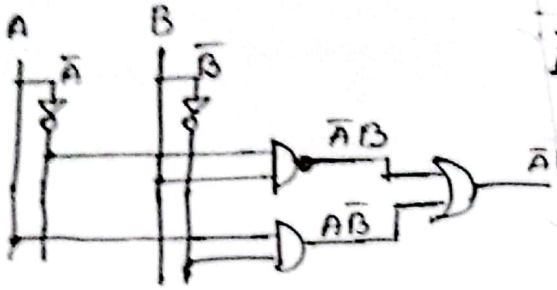
Logical Code

* ৫ NOR দিয়ে X-OR বাস্তবায়ন:

$\bar{A}B + A\bar{B} = A \oplus B$.

A	B	A⊕B
0	0	0
0	1	1 ✓
1	0	1 ✓
1	1	0

$\bar{A}B + A\bar{B}$.



$\bar{A}B + A\bar{B} = A \oplus B$.

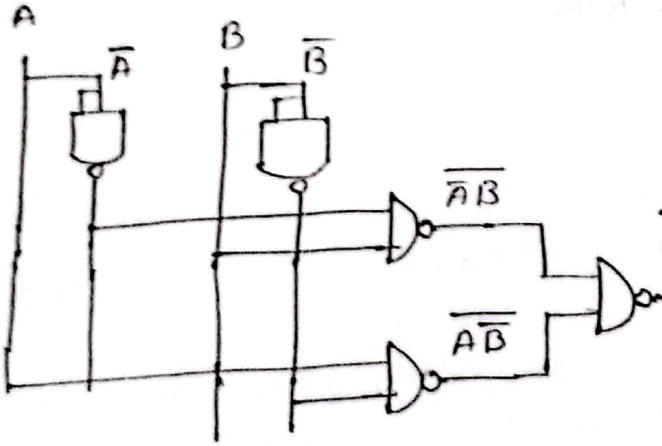
* NAND Gate দিয়ে:

$A \oplus B = \overline{\bar{A}B + A\bar{B}}$

$= \overline{\overline{\bar{A}B + A\bar{B}}}$

$= (\overline{\bar{A}B})(\overline{A\bar{B}})$

$\overline{A+B} = \bar{A} \cdot \bar{B}$

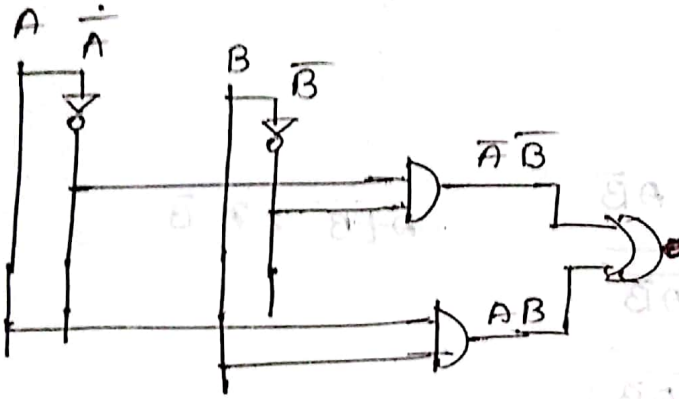


$(\overline{\bar{A}B})(\overline{A\bar{B}}) = \bar{A}B + A\bar{B} = A \oplus B$.

* মৌলিক গেট দিয়ে X-NOR Gate:

A	B	$\overline{A \oplus B}$
0	0	1
0	1	0
1	0	0
1	1	1

$\bar{A}B + A\bar{B} = \overline{A \oplus B}$

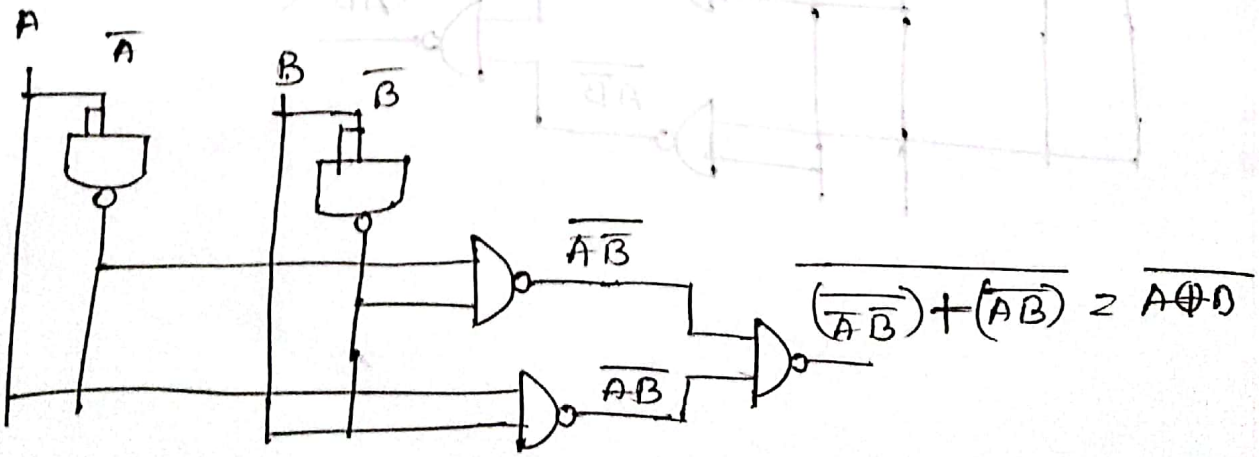


$\bar{A}B + A\bar{B} = \overline{A \oplus B}$.

* NAND Gate দিয়ে X-NOR

$$\overline{\overline{\bar{A}B + A\bar{B}}}$$

$$= \overline{(\overline{\bar{A}B}) \cdot (\overline{A\bar{B}})}$$



Logic Function simplification.

Ex-1: $\overline{(A+B+\bar{C})} \cdot B\bar{C}$

$= \bar{A}\bar{B}C \cdot B\bar{C}$ [De Morgan's]

$= \bar{A}\bar{B}B C \bar{C}$

$= \bar{A} \cdot 0 \cdot 0$

$= 0$

Ex-2 : $\bar{A}\bar{B}(\bar{A}+B)$

$= (\bar{A}+\bar{B})(\bar{A}+B)$

$= \bar{A}\bar{A} + \bar{A}B + \bar{B}\bar{A} + \bar{B}B$

$= \bar{A} + \bar{A}B + \bar{A}\bar{B} + 0$

$= \bar{A} \cdot 1 + \bar{A}B + \bar{A}\bar{B}$

$= \bar{A}(1 + B + \bar{B})$

$= \bar{A} \cdot 1$

$= \bar{A}$

Ex-3: $ABC + A\bar{B}C + AB\bar{C} + AC$ → $A+A=A$

$= ABC + A\bar{B}C + AC$

$= AC(B+\bar{B}) + AC$

$= AC \cdot 1 + AC$

$= AC + AC$

$= C(A+\bar{A})$

$= C \cdot 1 = C$

Ex-4: $\bar{A}\bar{B}C + \bar{A}B\bar{C} + A\bar{B}\bar{C} + ABC = A \oplus B \oplus C$

LHS: $\bar{A}\bar{B}C + \bar{A}B\bar{C} + A\bar{B}\bar{C} + ABC$

$= \bar{A}(\bar{B}C + B\bar{C}) + A(\bar{B}\bar{C} + BC)$

$= \bar{A}(B \oplus C) + A\,\overline{B \oplus C}$ $\quad B \oplus C = X$

$= \bar{A}X + A\bar{X}$

$= A \oplus X$

$= A \oplus B \oplus C$

Ex-5: $\overline{A \oplus B} = \bar{A}\bar{B} + AB$

LHS $= \overline{A \oplus B}$ $\quad [A \oplus B = \bar{A}B + A\bar{B}]$

$= \overline{\bar{A}B + A\bar{B}}$

$= \overline{\bar{A}B} \cdot \overline{A\bar{B}}$

$= (\bar{\bar{A}} + \bar{B})(\bar{A} + \bar{\bar{B}})$

$= (A + \bar{B})(\bar{A} + B)$

$= A\bar{A} + AB + \bar{A}\bar{B} + B\bar{B}$

$= 0 + AB + \bar{A}\bar{B} + 0$

$= \bar{A}\bar{B} + AB$.

Ex-6

$A + \bar{A}B + \bar{A}\bar{B} = 1$

L.H.S $= A + \bar{A}B + \bar{A}\bar{B}$

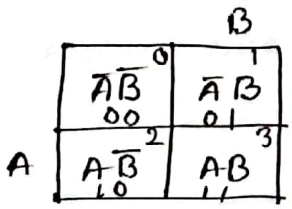
$= A + \bar{A}(B + \bar{B})$

$= A + \bar{A} \cdot 1$

$= A + \bar{A}$

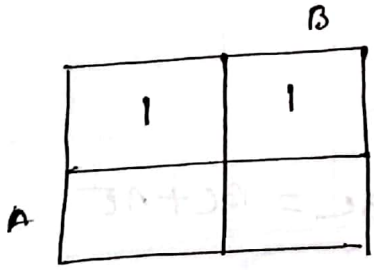
$= 1 = R.H.S.$

Karnaugh map

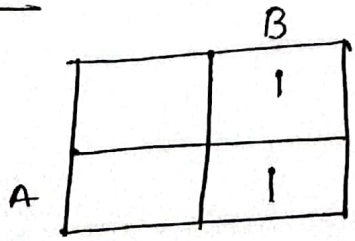


নিচের দুইটা বক্স A দ্বারা cover হবে।
উপরের " " $\bar{A}$ " " "
ডানদিকের বক্স B দ্বারা " "
বামদিকে " $\bar{B}$ " " "

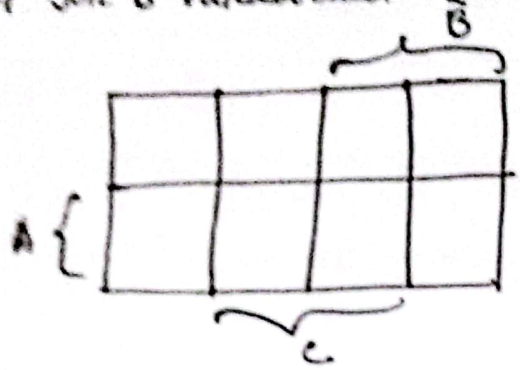
Ex-1: $\bar{A}\bar{B} + \bar{A}B = ? \rightarrow \bar{A}$

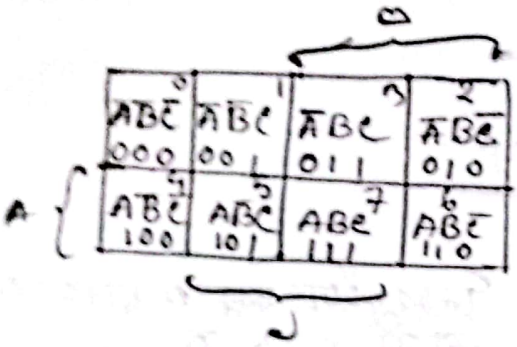


Ex-2: $AB + \bar{A}B = B$

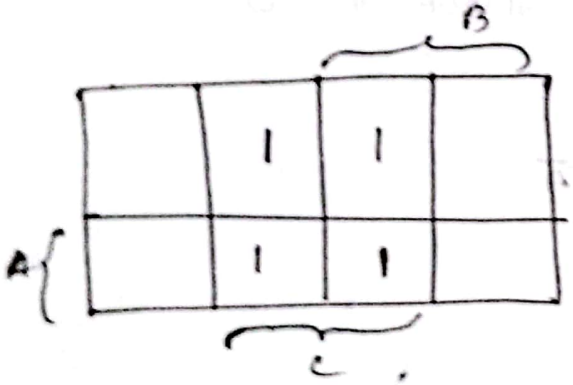


K-Map for 3 variables: $2^3 = 8$ box.

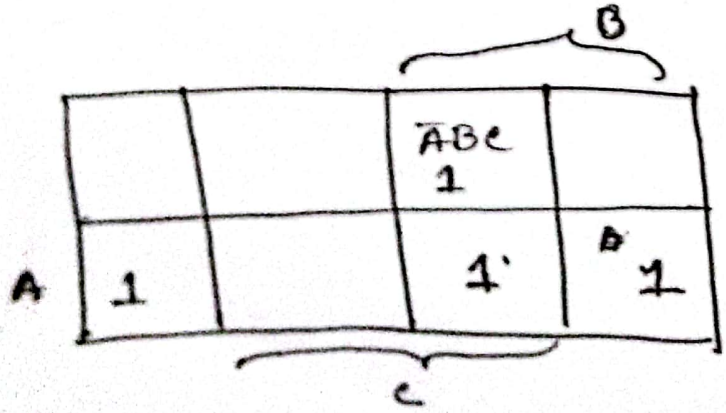




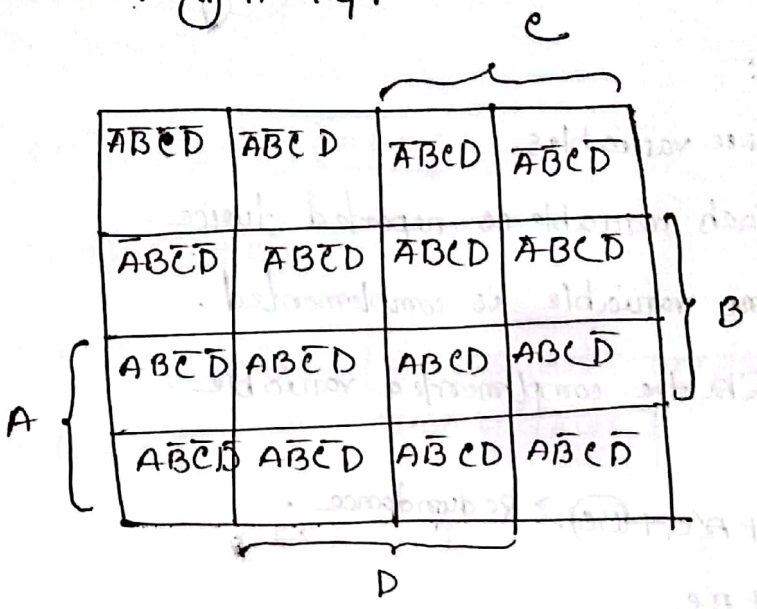
Ex-1 : $ABC + A\bar{B}C + \bar{A}BC + \bar{A}C = C$



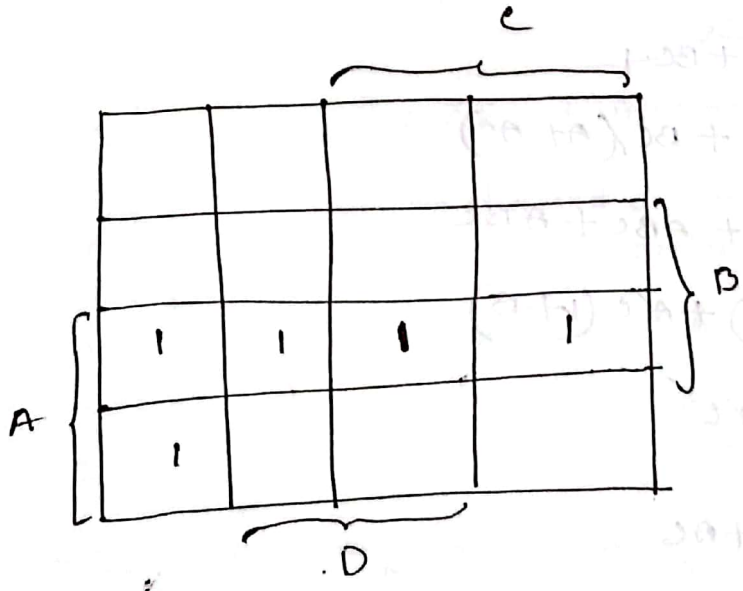
Ex-2 : $\bar{A}BC + A\bar{B}C + AB\bar{C} + ABC = BC + AC$



* 4টি Variable এর জন্য K-map:



Ex-1: $ABCD + AB\bar{C}\bar{D} + \bar{A}B\bar{C}\bar{D} + AB\bar{C}D + AB\bar{C}\bar{D} = AB + A\bar{C}\bar{D}$



* Consensus Theorem / Redundancy theorem.

Conditions:

(i) Three variables.

(ii) Each variable is repeated twice.

(iii) One variable is complemented.

(iv) Take the complimented variable.

$Y = AB + A'C + \overline{BC}$ → Redundance.

$= AB + A'C$

$Y = AB + A'C + BC$

$= AB + A'C + BC \cdot 1$

$= AB + A'C + BC(A + A')$

$= AB + A'C + ABC + A'BC$

$= AB(1 + C) + A'C(1 + B)$

$Y = AB + A'C$

(i) $F = A\overline{B} + B\overline{C} + AC$

$= B\overline{C} + AC$

(ii) $F = A\overline{B} + BC + \overline{AC}$

$= A\overline{B} + BC$

(iii) $(A+B)(\bar{A}+C)(B+C)$

$= (A+B)(\bar{A}+C)$

(iv) $F = (A+B).(\bar{B}+C)(A+C)$

$= (A+B)(\bar{B}+C)$

(v) $F = \bar{A}\bar{B} + A\bar{C} + \bar{B}\bar{C}$ (যেই অবস্থানেই complemented থাকে তবে complement হারত না নিবো।)

$= \bar{A}\bar{B} + A\bar{C}$

$0 \rightarrow \bar{A}$
$1 \rightarrow A$

Sum of Products

	A	B	C	F
0 m_0	0	0	0	0
1 m_1	0	0	1	0
2 m_2	0	1	0	1
3 m_3	0	1	1	0
4 m_4	1	0	0	1
5 m_5	1	0	1	1
6 m_6	1	1	0	1
7 m_7	1	1	1	1

→ sop form → for F=1
→ pos form

sop form

$F = \bar{A}B\bar{C} + AB\bar{C} + A\bar{B}C + ABC + ABC$

→ standard or canonical sop form
→ written directly from the truth table.

→ Minterm — $\bar{A}.B.\bar{C}$, AB, AC, etc

Maxterm — $(A+B+C)$

$F(A,B,C) = m_2 + m_4 + m_5 + m_6 + m_7$

$= \sum m(2,4,5,6,7)$

$F = \bar{A}B\bar{C} + A\bar{B}\bar{C} + A\bar{B}C + AB\bar{C} + ABC$

$= \bar{A}B\bar{C} + A\bar{B}(\bar{C}+C) + AB(\bar{C}+C)$

$= \bar{A}B\bar{C} + A\bar{B}.1 + AB$

$= \bar{A}B\bar{C} + A(\bar{B}+B)$ ∴ $F = \bar{A}B\bar{C} + A$

$= A + B\bar{C}$ → Minimal sop form.

Product of sum form (POS)

→ Pos form is used when the output is 0 or low.

0 → A
1 → $\bar{A}$

A	B	C	Y
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	1

$Y = (A+B+C).(A+B+\bar{C}).(A+\bar{B}+\bar{C})$ maxterm

$\bar{Y} = \bar{A}\bar{B}\bar{C} + \bar{A}BC + A\bar{B}C$ sop ⇌ pos

$\overline{\bar{Y}} = \overline{\bar{A}\bar{B}\bar{C} + \bar{A}BC + A\bar{B}C}$

$= (\overline{\bar{A}\bar{B}\bar{C}}).(\overline{\bar{A}BC}).(\overline{A\bar{B}C})$

$Y = \underbrace{(A+B+C)}_{X}.\underbrace{(A+B+\bar{C})}_{X}.(A+\bar{B}+\bar{C})$ $(A+B)(A+C) = A+BC$

$= (X + C.\bar{C})(A+\bar{B}+\bar{C})$

$= (A+B)(A+\bar{B}+\bar{C})$

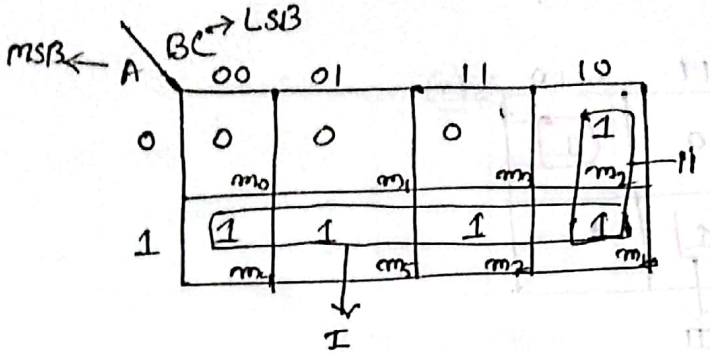
$= A + B.(\bar{B}+\bar{C})$

$= A + B\bar{B} + B\bar{C}$

$= A + B\bar{C}$

$= (A+B)(A+\bar{C})$ → minimal pos form.

* Extends K- Map:



Pairing

2 → 1's
4 → 1's
8 → 1's
16 → 1's.

$F = I + II$

group of 1's (Implicants)

$F = A \cdot 11 + B\bar{C}$

K map maintain $(A + \bar{A}) = 1$ rules, therefore we don't take variable which is changing.

0	$\bar{A}$
1	A

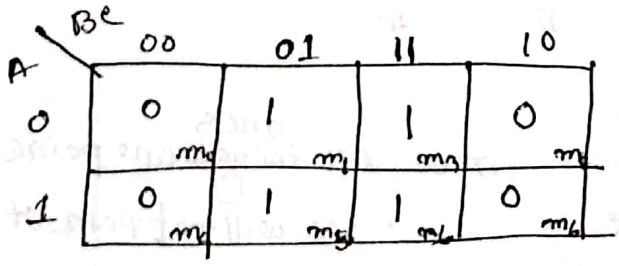
(Ex-1): $f(A,B,C) = \Sigma m(1,3,5,7)$

(i) Find out number of variables;

$n = 3$ (A, B, C)

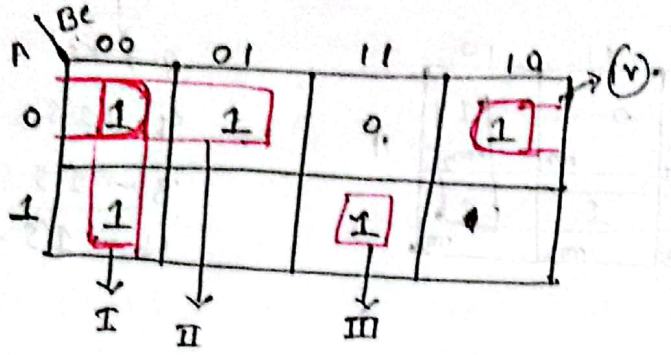
(ii) Find out number of cells in k'map.

$2^n = 2^3 = 8$ cells.



$F = C$

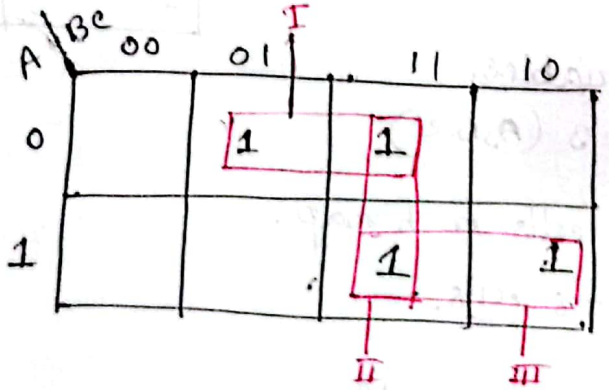
(ii) $F(A,B,C) = \Sigma m\ (0,1,2,4,7)$



$F = I + II + III + IV$

$= \bar{B}\bar{C} + \bar{A}\bar{B} + ABC + \bar{A}\bar{C}$

(ii) $F(A,B,C) = \Sigma m\ (1,3,6,7)$



→ NOT changing.

$F = I + II + III$

once All ones are paired

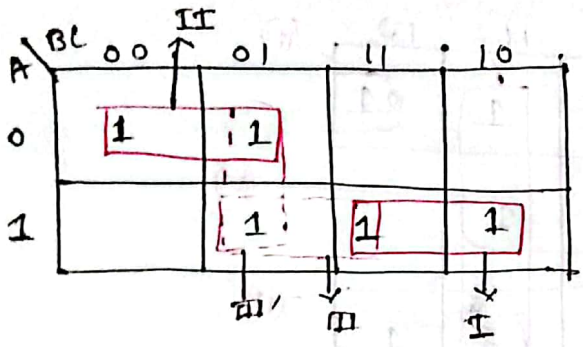
we will not pair it again

$= \cancel{C + BC + C}$

$= AB + \bar{A}C + BC$ Redundance

$= AB + \bar{A}C$

(Ex-iv): $F(A,B,C) = \Sigma m(0,1,5,6,7)$



$F = I + II + III$

$= A\bar{B} + \bar{A}\bar{B} + AC$

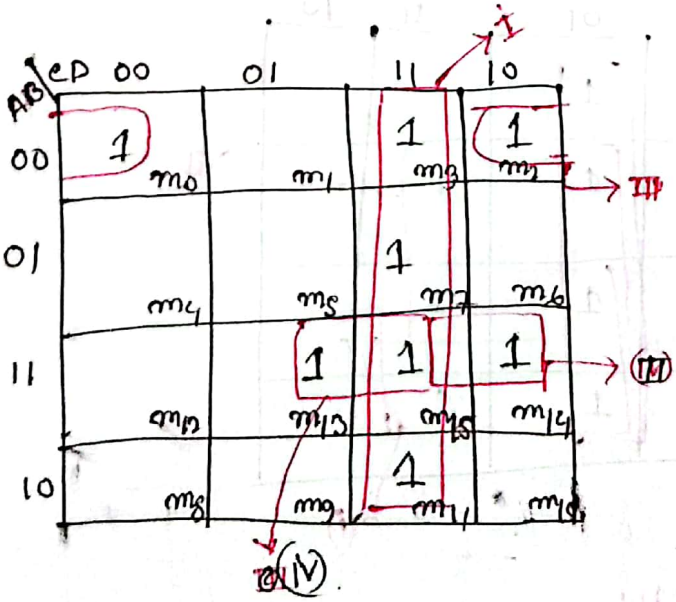
$F = I + II + III'$

$= A\bar{B} + \bar{A}\bar{B} + \bar{B}C$

Note The result is minimum but may not be the same.

* K'Map with 4 variables :-

(Ex-i): $F(A,B,C,D) = \Sigma m(0,2,3,7,11,13,14,15)$



combines

4 → 2 variables reduced

2 → 1 " "

8 → 3 " "

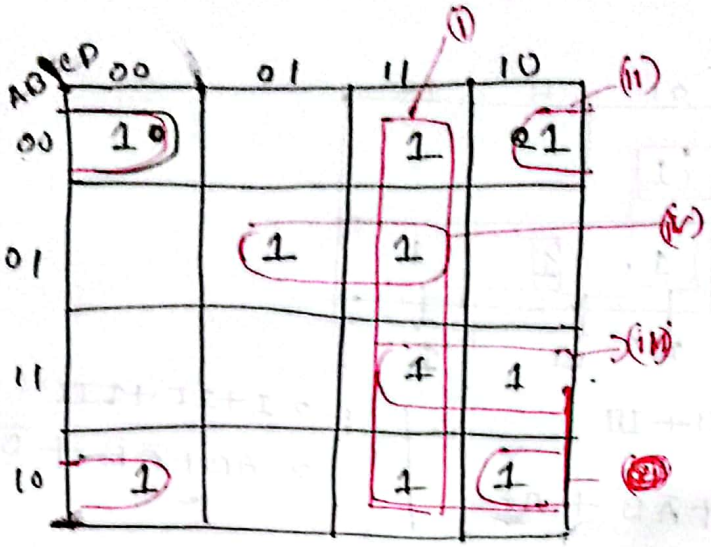
16 → 4 " "

$F = I + II + III + IV$

$= CD + A\bar{B}\bar{D} + ABC + ABD$

CamScanner

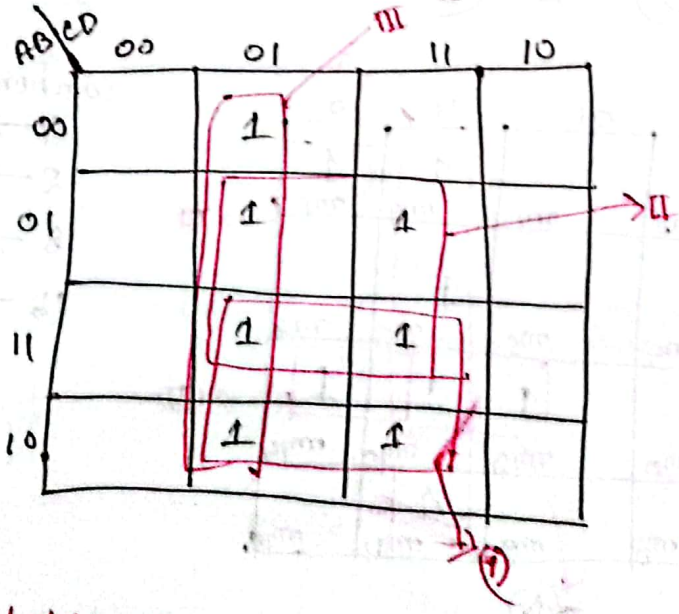
* $F(A,B,C,D) = \Sigma m(0,2,3,5,7,8,10,11,14,15)$



$F = I + II + III + (iv) + (v)$

$= CD + \bar{B}D + AC + \bar{A}BD$

Ex- $F(x,y,z,w) = \Sigma m(1,5,7,9,11,13,15)$



$F = I + II + III$

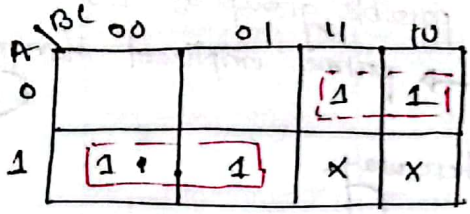
$= xw + yw + zw.$

* Don't care in K-map:

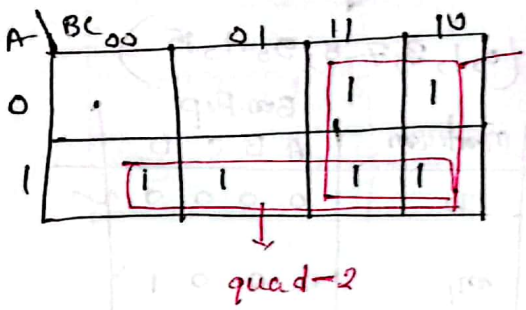
$F(A,B,C) = \Sigma m(2,3,4,5) + \Sigma d(6,7)$. (d → don't care)

d = 1 → min-term

→ 0 → max-term.



$F = \bar{A}B + A\bar{B} = A \oplus B$.



$F = \bar{B} + A$

Implicants → Group of one's is called implicants.

prime implicants → The largest group of 1's.

essential prime implicants → At least there is ~~single~~ simple 1's which cannot be combined in any other way.

Quine – McCluskey Minimization Technique

1. Prime implicant. → largest possible group of 1's.
2. Essential prime implicants. → prime implicant having one minterm that can not combined any other way.

Ex-1: $Y(A,B,C,D) = \Sigma m(0,1,3,7,8,9,11,15)$

A B C D
0 – 0 0 0 0
1 – 0 0 0 1
3 – 0 0 1 1
7 – 0 1 1 1
8 – 1 0 0 0
9 – 1 0 0 1
11 – 1 0 1 1
15 – 1 1 1 1

step-1:

Group	Minterm	Bin. Rep A B C D	
0	m_0	0 0 0 0	✓
1	m_1	0 0 0 1	✓
	m_8	1 0 0 0	✓
2	m_3	0 0 1 1	✓
	m_9	1 0 0 1	✓
3	m_7	0 1 1 1	✓
	m_{11}	1 0 1 1	✓
4	m_{15}	1 1 1 1	✓

step-2:

Group	Matched pairs.	Bin Rep A B C D	
0	$m_0 - m_1$	0 0 0 –	✓
	$m_0 - m_8$	– 0 0 0	✓
1	$m_1 - m_3$	0 0 – 1	✓
	$m_1 - m_9$	– 0 0 1	✓
	$m_8 - m_9$	1 0 0 –	✓
2	$m_3 - m_7$	0 – 1 1	✓
	$m_3 - m_{11}$	– 0 1 1	✓
	$m_9 - m_{11}$	1 0 – 1	✓
3	$m_7 - m_{15}$	– 1 1 1	✓
	$m_{11} - m_{15}$	1 – 1 1	✓

step-3:

Group	matched pairs	B.R A	B	C	D	
0	$m_0-m_1-m_8-m_9$	—	0	0	—	$\bar{B}\bar{C}$
	$m_0-m_8-m_1-m_9$	—	0	0	—	
1	$m_1-m_3-m_9-m_{11}$	—	0	—	1	$\bar{B}D$
	$m_1-m_9-m_3-m_{11}$	—	0	—	1	
2.	$m_3-m_7-m_{11}-m_{15}$	—	—	1	1	CD
	$m_3-m_{11}-m_7-m_{15}$	—	—	1	1	

$\bar{B}\bar{C}$, $\bar{B}D$, CD → Prime Implicants

P.I	minterms involves	0	1	3	7	8	9	11	15
$\bar{B}\bar{C}$	0, 1, 8, 9	⊗	×			⊗	×		
$\bar{B}D$	1, 3, 9, 11		×	×			×	×	
CD	3, 7, 11, 15			×		⊗		×	⊗

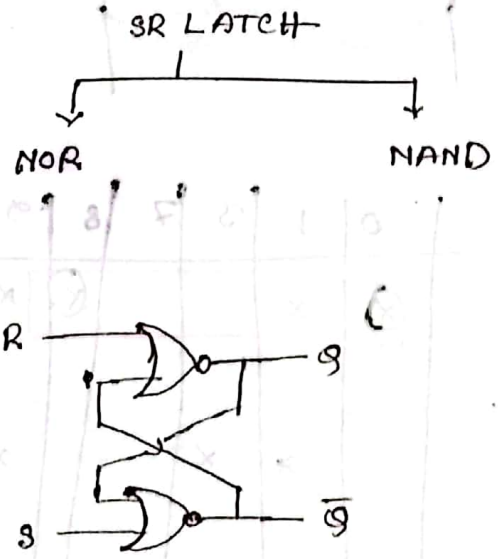
$Y = \bar{B}\bar{C} + CD$

Sequencial Circuit

In the circuits where the present output depends on the past output.

SR LATCH

* The basic storage element is called latch. As the name suggests it latches 0 or 1.



R = Reset $Q = 0$
S = set $Q = 1$

* Truth Table for 2 input NOR Gate.

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	0

(যেকোন input 1 হলেই output 0.

case-1: $S=0, R=1, Q=0, \bar{Q}=1$.

$S=0, R=0, Q=0, \bar{Q}=1$ → memory.

Case-2: $S=1, R=0, Q=1, \bar{Q}=0$

$S=0, R=0, Q=1, \bar{Q}=0$

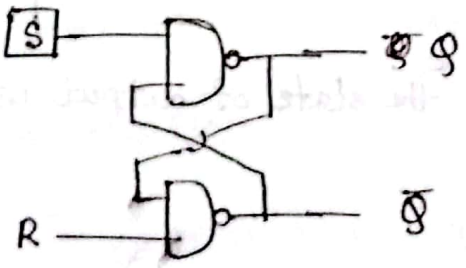
Case-3: $S=1, R=1, Q=0, \bar{Q}=0$
$S=0, R=0, Q=0, \bar{Q}=1$
$Q=1, \bar{Q}=0$
→ NOT used in SR LATCH. invalid state.

S	R	Q	$\bar{Q}$
0	0	memory (as before)	
0	1	0	1
1	0	1	0
1	1	NOT used	

→ Truth Table for SR-NOR LATCH.

* NAND LATCH:

→ Different between NOR and NAND LATCH.



S	R	Q	$\bar{Q}$
0	0	NOT used	
0	1	1	0
1	0	0	1
1	1	memory state	

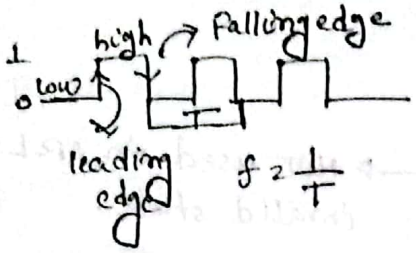
A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

সকেশন ইনপুট 0 হলেই output 1.

C-2 $S=0, R=1, Q=1, \bar{Q}=0$
$S=1, R=1, Q=1, \bar{Q}=0$ — memory

C-2 $S=1, R=0, Q=0, \bar{Q}=1$
$S=1, R=1, Q=0, \bar{Q}=1$ — (memory)

clock



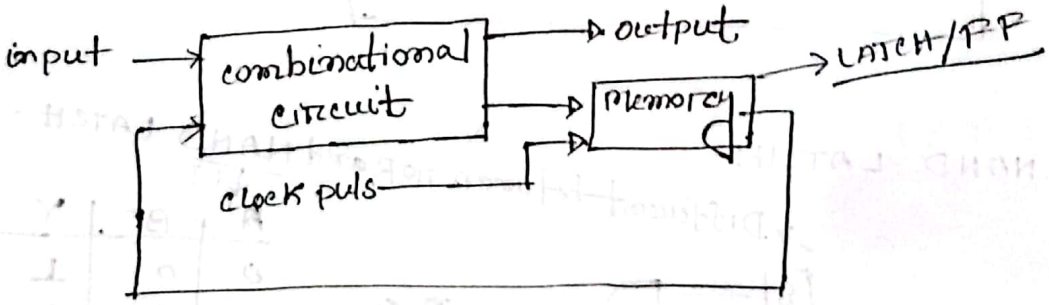


→ only work on high state.

Duty cycle → Ratio of time $\frac{\text{signal} \uparrow}{\text{Total time}} = \frac{t/2}{t}$

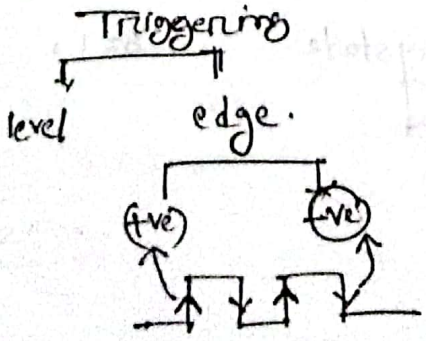
$= \frac{1}{2} = 50\%$.

Triggering Method



input is given and the state of output is stored in the memory.

The clock pulse changes the memory state stored previously. After new input is given it will not change the stored output untill the clock puls is given.

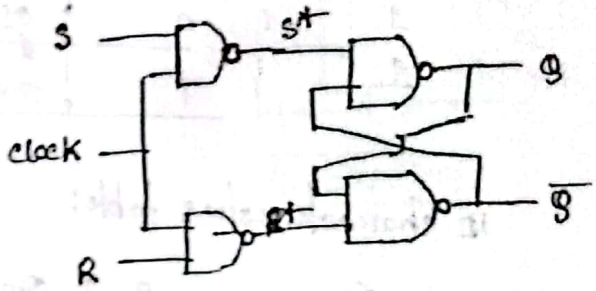


* LATCH is level sensitive. It will work on enable state.
* FlipFlop is operational on eg edge Triggering.

SR FlipFlop

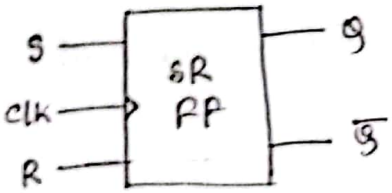
* SR HAND LATCH. T.T.

S*	R*	Q	$\overline{Q}$
0	0	Not used	
0	1	1	0
1	0	0	1
1	1	Memory	



$S^* = \overline{S.clk} = \overline{S} + \overline{clk}$

$R^* = \overline{R.clk} = \overline{R} + \overline{clk}$



edge Triggered.

$S^* = \overline{S} + 1 = 1$
$R^* = R + 1 = 1$
} clock = 0.

$S^* = \overline{S} + 0 = \overline{S}$
$R^* = \overline{R} + 0 = \overline{R}$
} clock = 1.

T.T for SR FlipFlop:

clk	S	R	Q	$\overline{Q}$	
0	x	x	memory		→ clock 0 Then input doesn't matters.
1	0	0	Memory		
1	0	1	0	1	
1	1	0	1	0	
1	1	1	Not used.		

* Truth Table for SR FlipFlop:

CLK	S	R	Q_{n+1}
0	X	X	Q_n
1	0	0	Q_n
1	0	1	0
1	1	0	1
1	1	1	invalid

Q_{n+1} = Future state (next)

Q_n = present state

* characteristics table:

Q_n	S	R	Q_{n+1}
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	X
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	X

changing अवस्था X

* Excitation table:

Q_n	Q_{n+1}	S	R
0	0	0	X
0	1	1	0
1	0	0	1
1	1	X	0

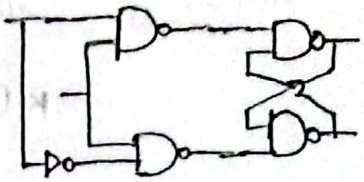
Q_n \ SR	00	01	11	10
0	0	0	X	1
1	1	0	X	1

(i)

(ii)

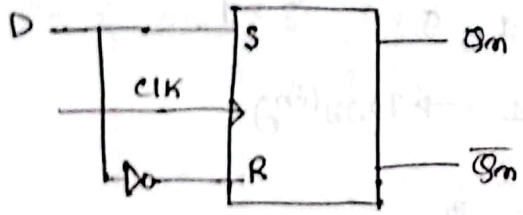
$Q_{n+1} = i + ii$

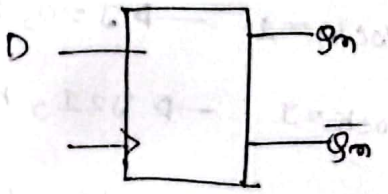
$= S + Q_n\bar{R}$



D-flipflop

D – Flip Flop





Truth Table:

clk	D	Q_{n+1}
0	X	Q_n
1	0	0
1	1	1

Excitation Table:

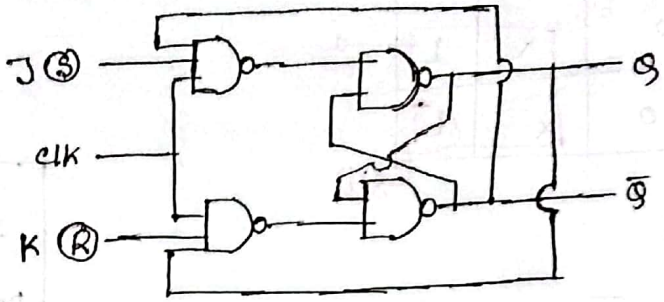
Q_n	Q_{n+1}	D
0	0	0
0	1	1
1	0	0
1	1	1

characteristics Table:

Q_n	D	Q_{n+1}
0	0	0
0	1	1
1	0	0
1	1	1

$Q_{n+1} = D$

J-K Flip Flop



clock = 0 → memory state.

clock = 1 → J = 1, K = 0, Q = 1, $\bar{Q}$ = 0,

clock = 1 → J = 0, K = 1, Q = 0, $\bar{Q}$ = 1.

clock = 1 → J = 1, K = 1 → Toggling.

T.T

clk	J	K	Q_{n+1} $\bar{Q}$
0	✗	✗	memory (Q_n)
1	0	0	Q_n
1	0	1	0
1	1	0	1
1	1	1	Toggle ($\bar{Q}_n$)

ch. Table:

Q_n	J	K	Q_{n+1}
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	0

* Excitation Table:

Q_n	Q_{n+1}	J	K
0	0	0	x
0	1	1	x
1	0	x	1
1	1	x	0

For J

Q_n \ Q_{n+1}		
		1
	x	x

$J = Q_{n+1}$

For K

Q_n \ Q_{n+1}		
	x	x
	1	0

$K = \overline{Q_{n+1}}$

* From ch. Table:

Q_n \ JK	00	01	11	10
0	0	0	1	1 (i)
1	1	0	0	1 (ii)

$$Q_{n+1} = \overline{Q_n}\,J + Q_n\overline{K}$$

* Condition to overcome Racing:

Not used (i) T/2 of clock < prop. delay

(ii) edge triggering.

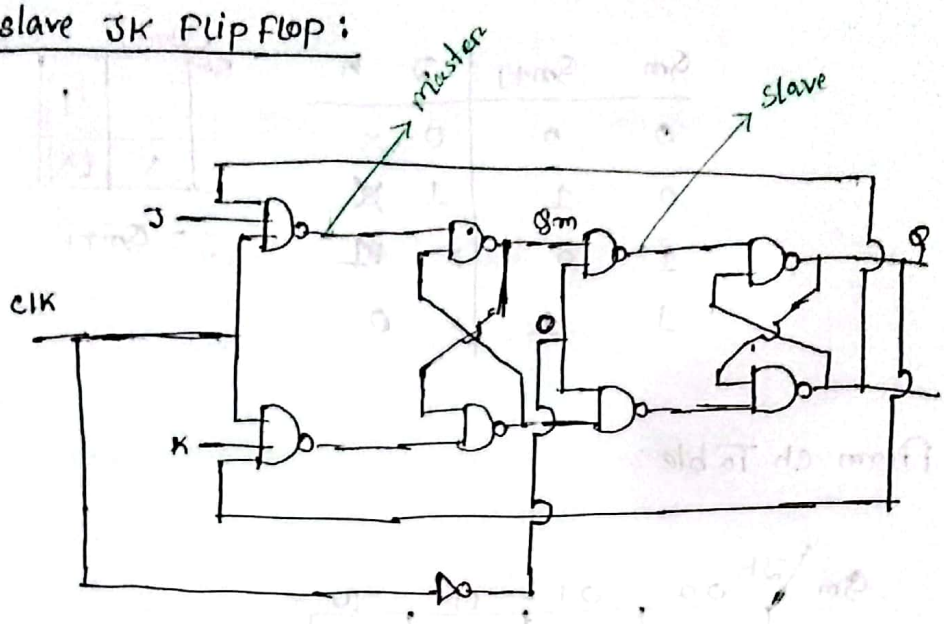
(iii) Master-slave (m.s is same as negative edge triggering)

* flipflops takes some times to generate the outputs

changing of output continuously is called racing.

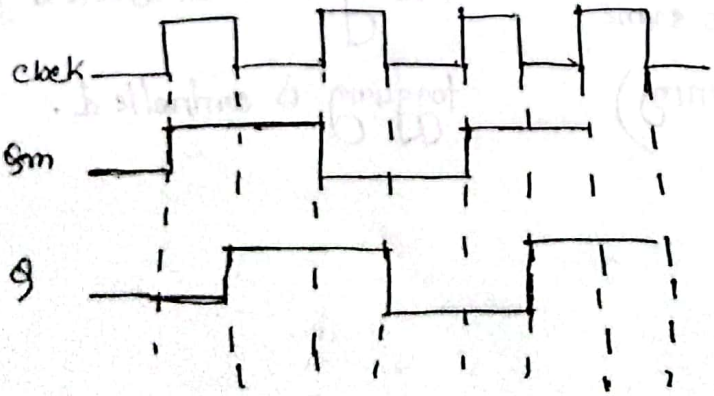
racing is uncontrolled whereas toggling is controlled.

* Master slave JK Flip flop :



* When clock is high master is operational and slave is kept on memory stage.

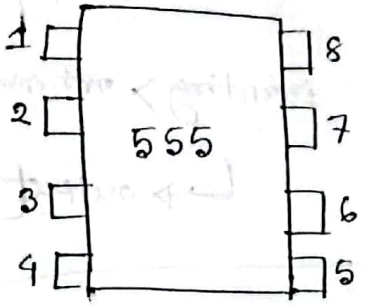
Instead of changing the output continuously the output changes once in a clock cycle. Cause the effect of feedback is eliminated.



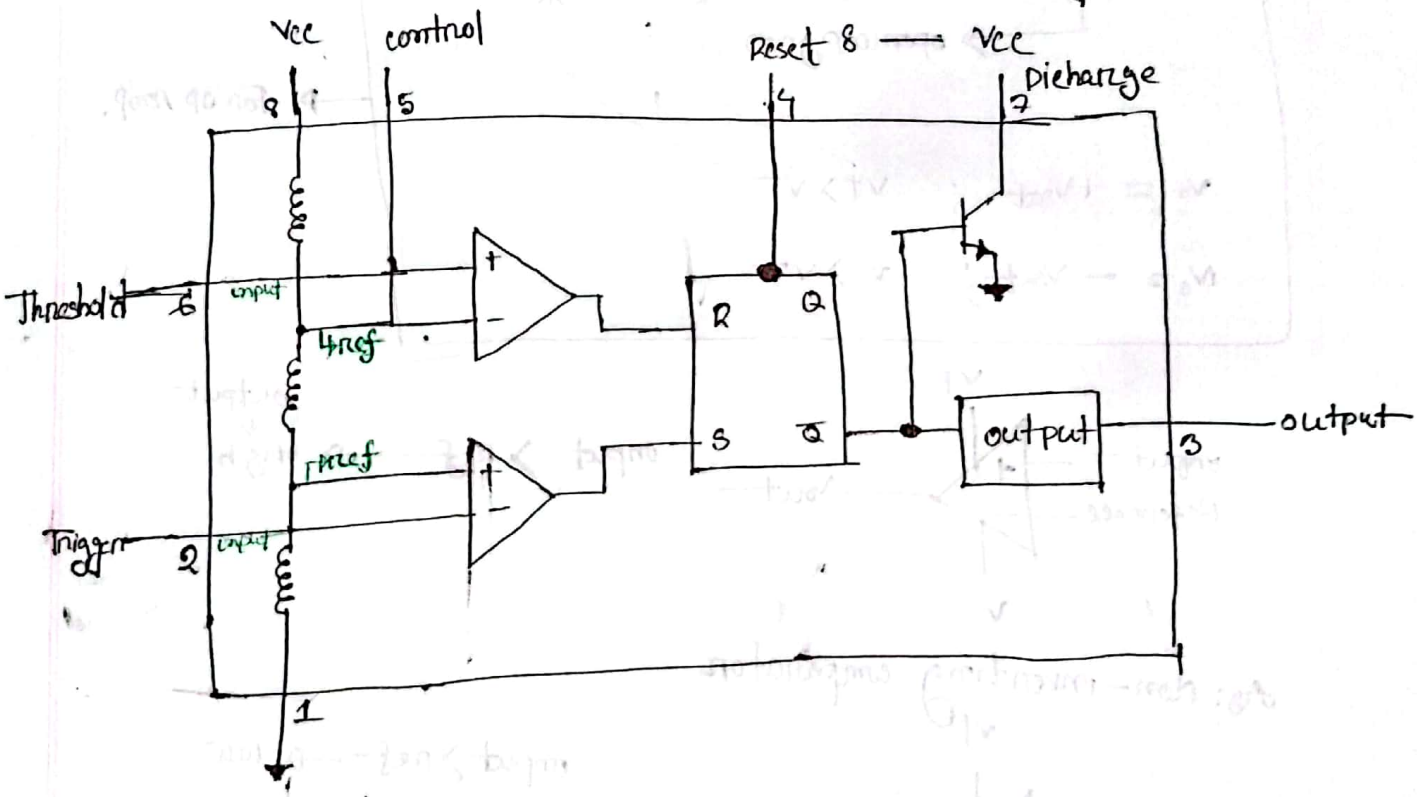
clock	J	K	Q_{n+1}
1	1	1	$\overline{Q}$

Toggling

555 Timer

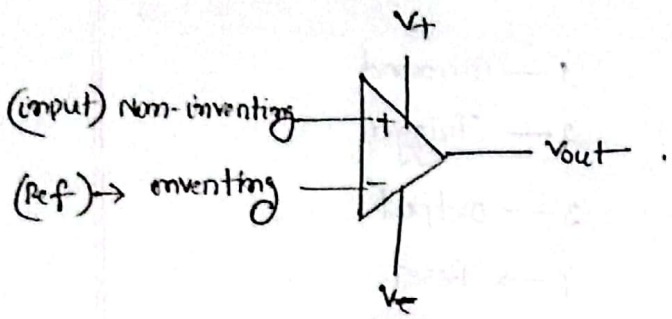


1 — Ground
2 — Trigger
3 — output
4 → Reset.
5 — control
6 — Threshold.
7 — Discharge.
8 — Vcc



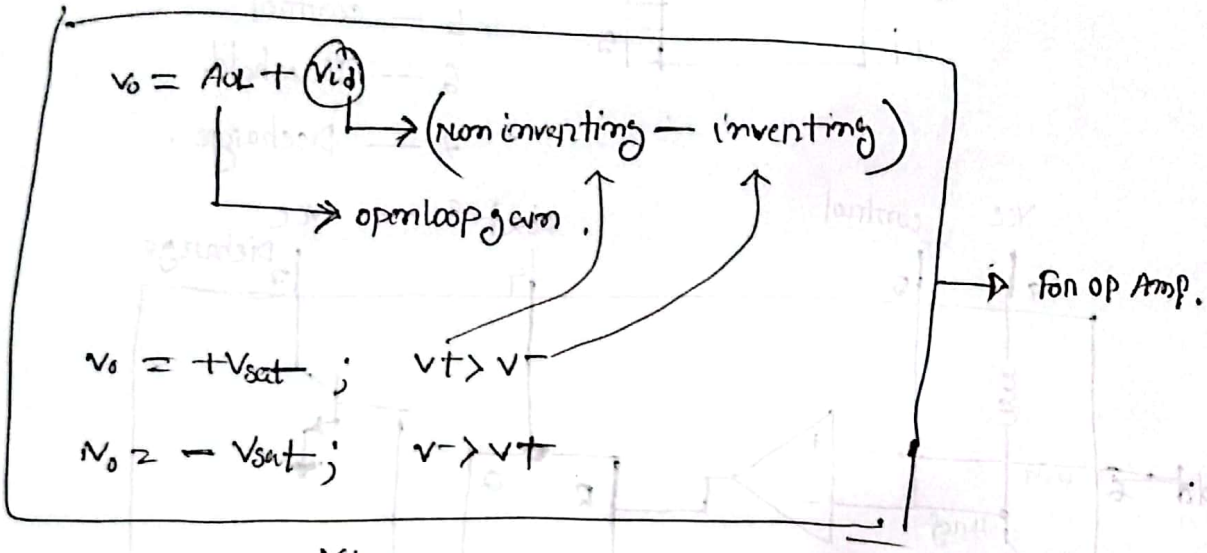
$\frac{2R \times Vcc}{2R+R} = \frac{3}{2} Vcc$ → voltage at 1st comparator

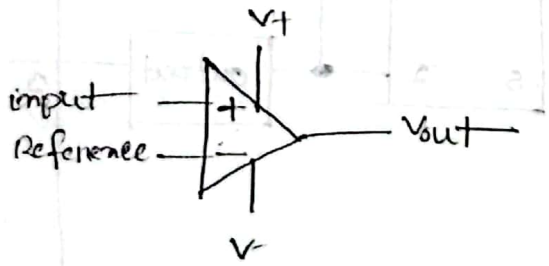
$\frac{R}{R+2R} Vcc = \frac{1}{3} Vcc$ → voltage at 2nd comparator



Non inverting input > inventing
↳ output → high

inventing > non inventing
↳ output → low.





input > Ref → output ↓ high

fig: Non-inverting comparator

input > ref → low

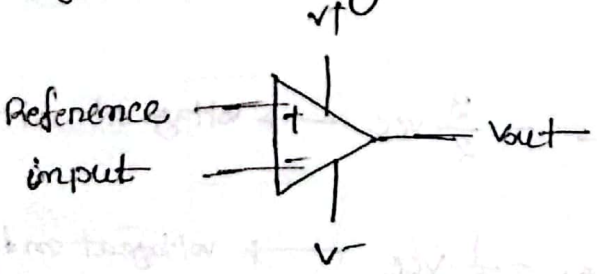
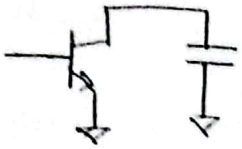
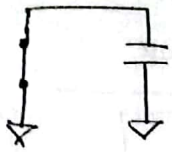
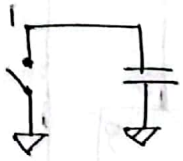


fig: inverting comparator.





→ when output is 0
then $\bar{Q} = 1$ and Transistor will act as closed switch.
capacitor will be discharged.



$\bar{Q} = 0$ output → 1
Transistor will act as open circuit

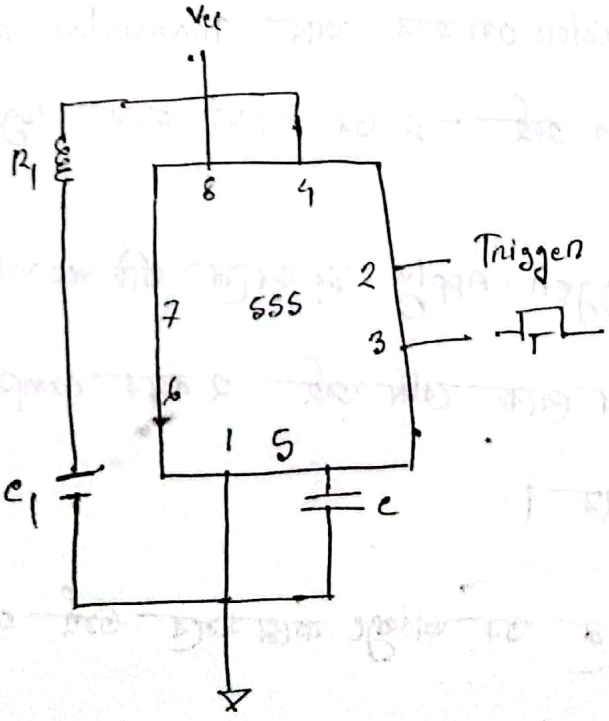
Monostable Multivibrator:

(i) Generating Timing delay.

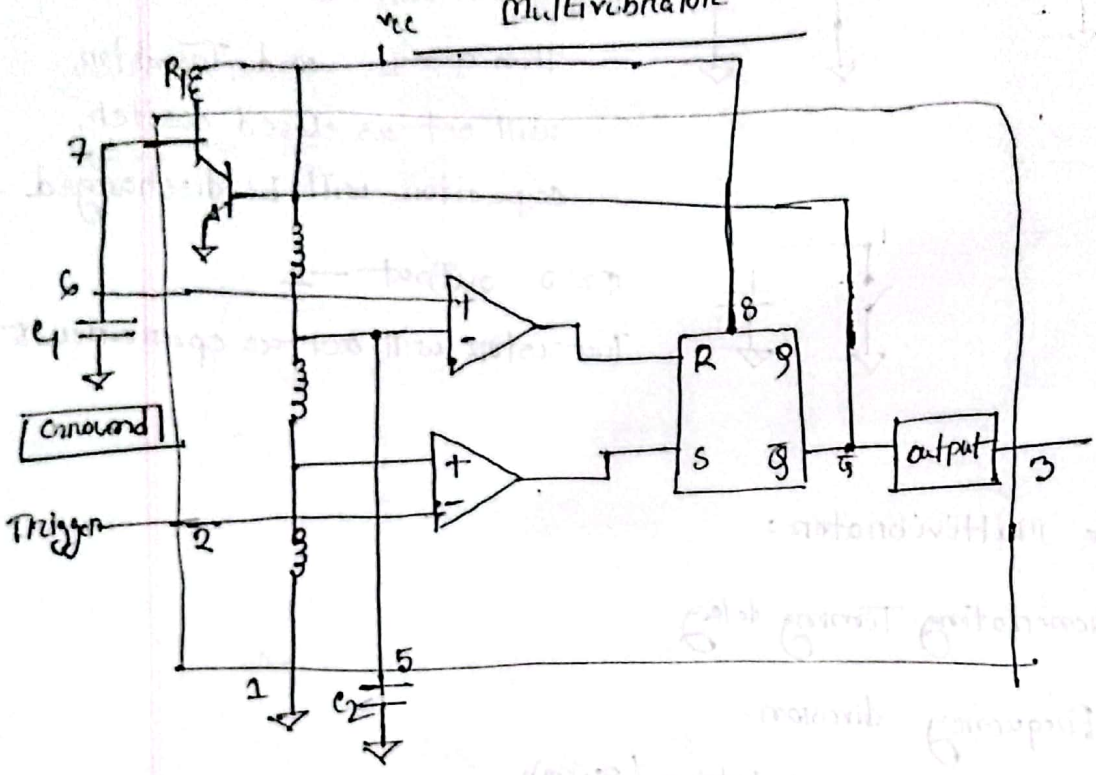
(ii) Frequency division.

(iii) pulse width modulation. (PWM)

(iv) switching the relay.



555 Timer as Monostable Multivibrator



$\bar{Q}$ = 1 output = 0 হলে $\bar{Q}$ = 1

$\bar{Q}$ = 1 হলে Transistor on হবে আর Transistor থেকে 6 → ground করবে তাই — 6 এর input হবে logic 0 (low)

Trigger Apply না করলে এটি Vcc voltage এর থাকবে যা রেফারেন্স থেকে বেশি তাই 2 নম্বর comparator এর আউটপুট 0 হবে।

Trigger Apply করলে এর voltage কমে যাবে তাই output 1 হবে।

$t = 0^+$ Transistor will be switched off. And the capacitor will start charging. As soon as the voltage crosses the reference voltage it will produce logic 1 at first comparator.

- Time duration of the pulse: $\boxed{1.1 R_1 C_1 = T}$

uses: ① Generating Time delays ($T \geq 1.1 R_1 C_1$)

② Frequency division ($t > T$) → Triggering signal
→ output.

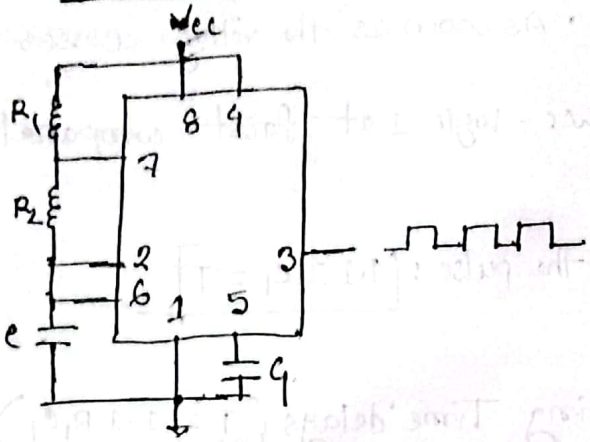
③ Pulse width modulation (Apply control signal)(AMod)
in this case the reference voltage will change continously based on modulation signal

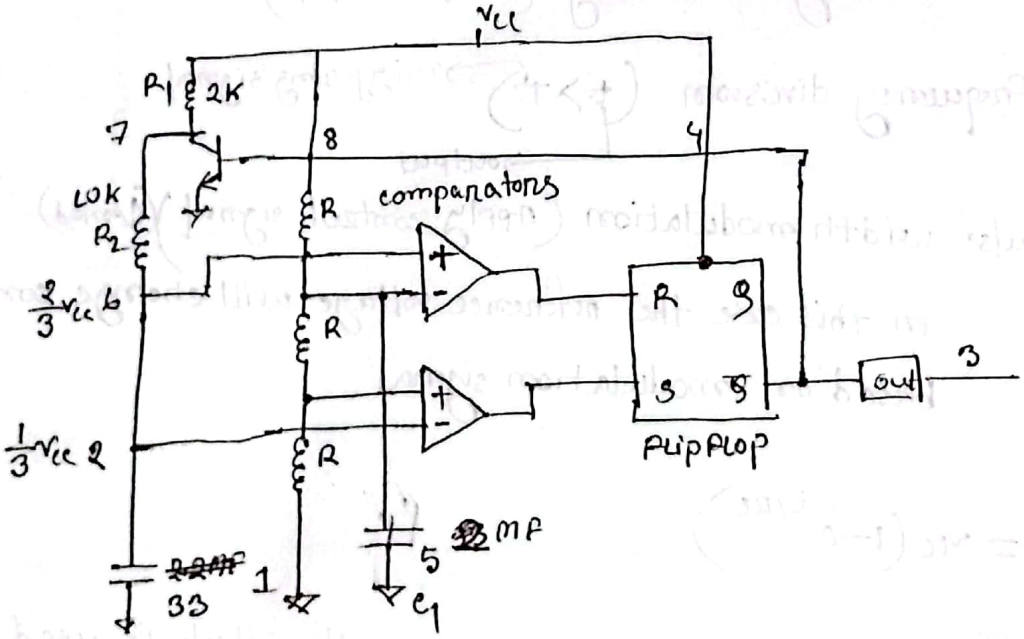
- $V_c(t) = V_{cc}\left(1 - e^{-t/Rc}\right)$

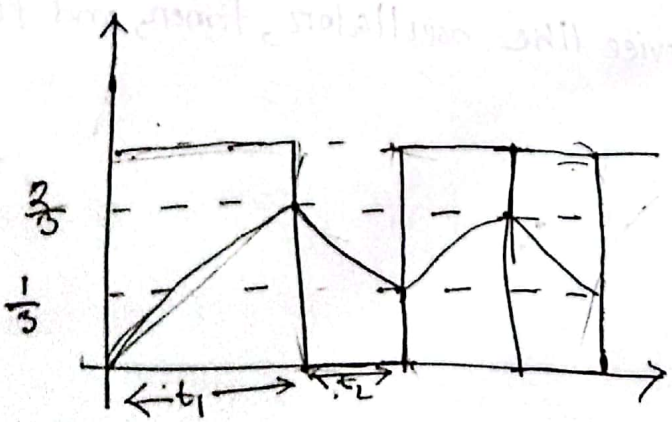
- The multivibrator is the electronic circuit which is used to implement two state device like oscillators, timers and Flipflop.



555 Astable multivibrator :







$t_1 > t_2$

- Charging → 2 Resistor
- discharging → 1 Resistor.

$t_1 = 0.693(R_1 + R_2)\,C$

$t_2 = 0.693\,R_2\,C$

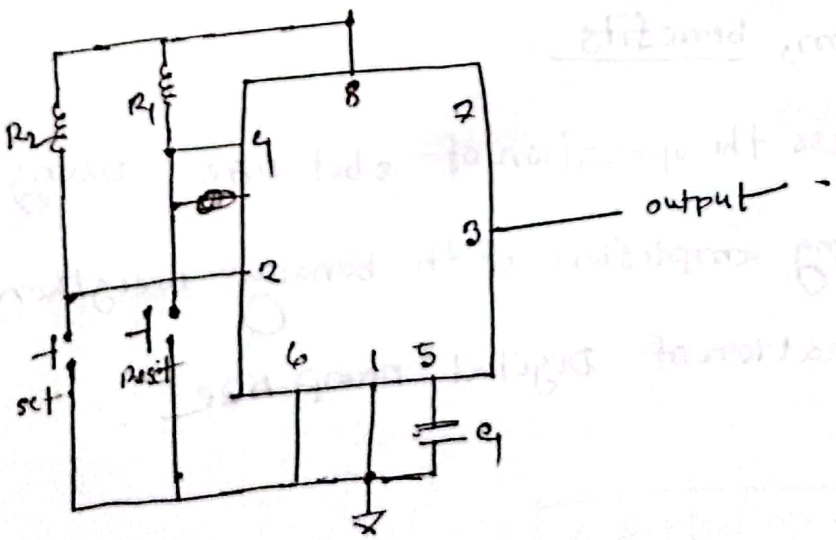
$T = t_1 + t_2 = 0.693\,(R_1 + 2R_2)\,C$.

Duty cycle $= \frac{t_1}{T}$.

$$= \frac{0.693\,(R_1 + R_2)}{0.693\,(R_1 + 2R_2)}$$

Duty cycle $= \frac{R_1 + R_2}{R_1 + 2R_2}$

Bistable multivibrator



DAC and ADC

Questions:

1. Resolution and accuracy of DAC.
2. Block Diagram of ADC and its basic operation.
3. DAC Related MATH.
4. block diagram of a successive approximation ADC and its basic operation.
5. basic operation of digital ramp ADC.
6. Transducer, Actuator.

*7. R/2R ladder DAC of 4 bit and write Vout expression, benefits

8. Discuss the operation of 8 bit DAC using op-amp summing amplifier with binary weighted resistor.
9. Operation of Digital ramp ADC.

TTL NOR

Analog signal ⟶ Transducer ⟶ Electrical signal

- Resolution = $\frac{\text{Full scale Range}}{2^{\text{No of bits}}} = \frac{V_{max} - V_{min}}{2^{\text{No of bits}}}$.

↓

How the quantized value is closed to the actual value.

The minimum change in the input signal which can be detected by ADC

or $\frac{V_{ref}}{2^n}$ n = no of bits

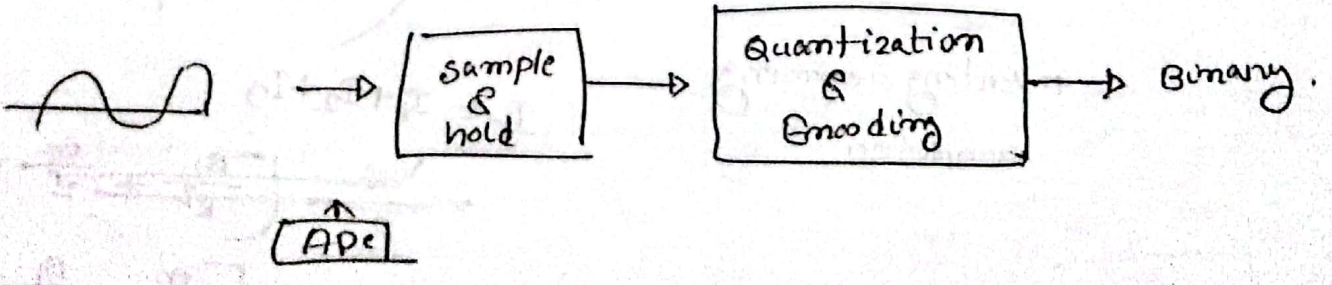
V_{ref} = ADC reference voltage.

- (3 bit range 0–1V)

$\frac{1}{2^3} = 0.125\,V$

That means $(0 - 0.125\,V)$ is considered as 0 cause the ADC will not be able to ~~conv~~ detect it.

- sampling rate $f_s = 2 \times f_{max}$



Binary weighted Resistor DAC

Resolution $= \frac{V_{ref} \times K}{2^n}$ $\quad K =$ internal gain of DAC

* Full scale output voltage (FSO) $= \frac{(2^3-1) \times V_{ref}}{2^3}$

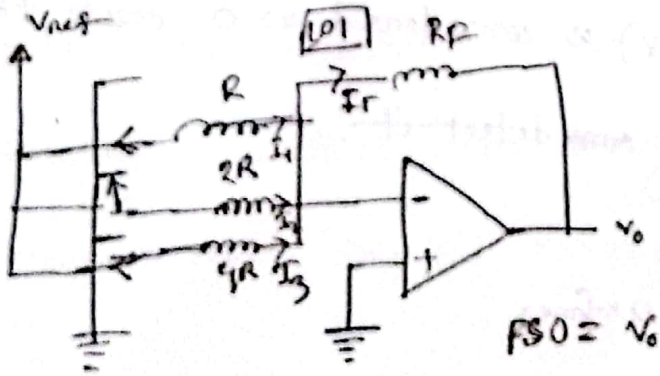
$= \frac{(2^n-1) \times V_{ref}}{2^n}$.

FSO $= (2^{n}-1) \times$ step size (Resolution)

step size $= \frac{FSO}{2^n-1} = \frac{V_{ref}}{2^n}$

FSO $=$ V_{ref} $-$ 1LSB.

$V_{out} = -\left(\frac{R_f}{R_1} V_1 + \frac{R_f}{R_2} V_2 + \frac{R_f}{R_3} V_3\right)$



inverting summing amplifier

$V_o = -R_f \times V_{ref}\left[\frac{B_2}{R} + \frac{B_1}{2R} + \frac{B_0}{4R}\right]$

$FSO = V_o = \left(\frac{-V_{ref} \times R_F}{R}\right) \sum_{i=0}^{i=N-1} \frac{B_i}{2^{(N-1)-i}}$

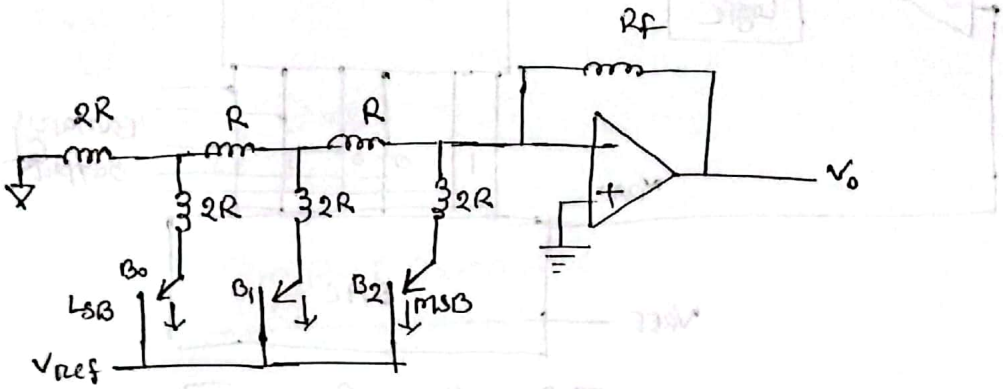
$I_T = I_1 + I_2 + I_3$

$= \frac{V_{ref}}{R}\left[\frac{B_2}{2^2} + \frac{B_1}{2^1} + \frac{B_0}{2^0}\right]$

$= \frac{V_{ref}}{R} \times \left[\frac{B_0}{2^2} + \frac{B_1}{2^1} + \frac{B_2}{2^0}\right]$

$$I_T = \left(\frac{V_{ref}}{R}\right) \sum_{i=0}^{i=N-1} \frac{B_i}{2^{(N-1)-i}}$$

R-2R Ladder DAC



Advantages of R-2R ladder DAC:

(i) uses only two values of Resistor — easy and accurate fabrication is possible.

(ii) Easily scalable to any desired number of bits.

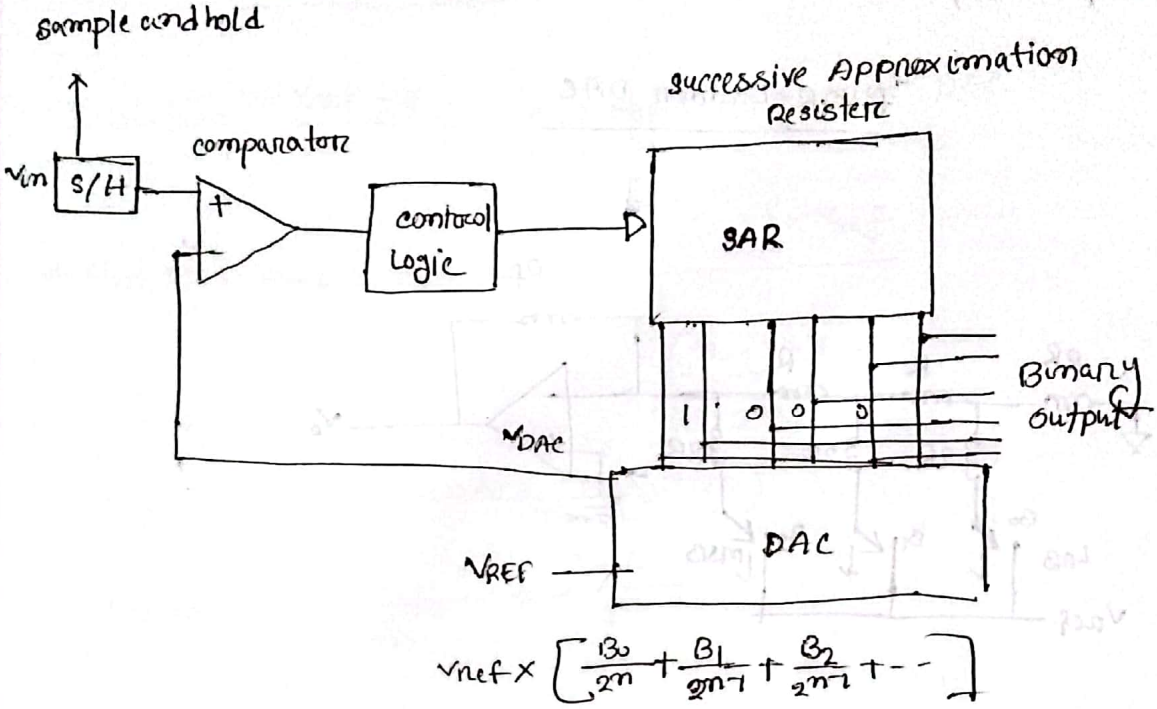
(iii) output impedance is R. Regardless of number of bits.

* output of a ~~R-2R~~ ladder $V_{out} = V_{ref}\left(\frac{B_0}{2^n} + \frac{B_1}{2^{n-1}} + \frac{B_2}{2^{n-2}} + \cdots + \frac{B_{n-2}}{2^2} + \frac{B_{n-1}}{2^1}\right)$

* Output of DAC $V_{out} = V_{ref}\left(-\frac{R_F}{R}\right)\left(\frac{B_0}{2^n} + \frac{B_1}{2^{n-1}} + \frac{B_2}{2^{n-2}} + \cdots\right)$

$$V_o = -\left(\frac{V_{ref} \times R_F}{R}\right) \times \sum_{i=0}^{i=N-1} \left(\frac{B_i}{2^{(N-i)}}\right)$$

Successive Approximation ADC



- whenever the new convention starts the SAR set the MSB to 1 and all other bits to zero.

if 1000 → $V_{in} < V_{DAC}$ MSB set to '0' and next bit set to 1 for the next comparison.

0100

$V_{in} > V_{DAC}$ 1000 (remains) and the next bit is set 1 for the next comparision.

1100

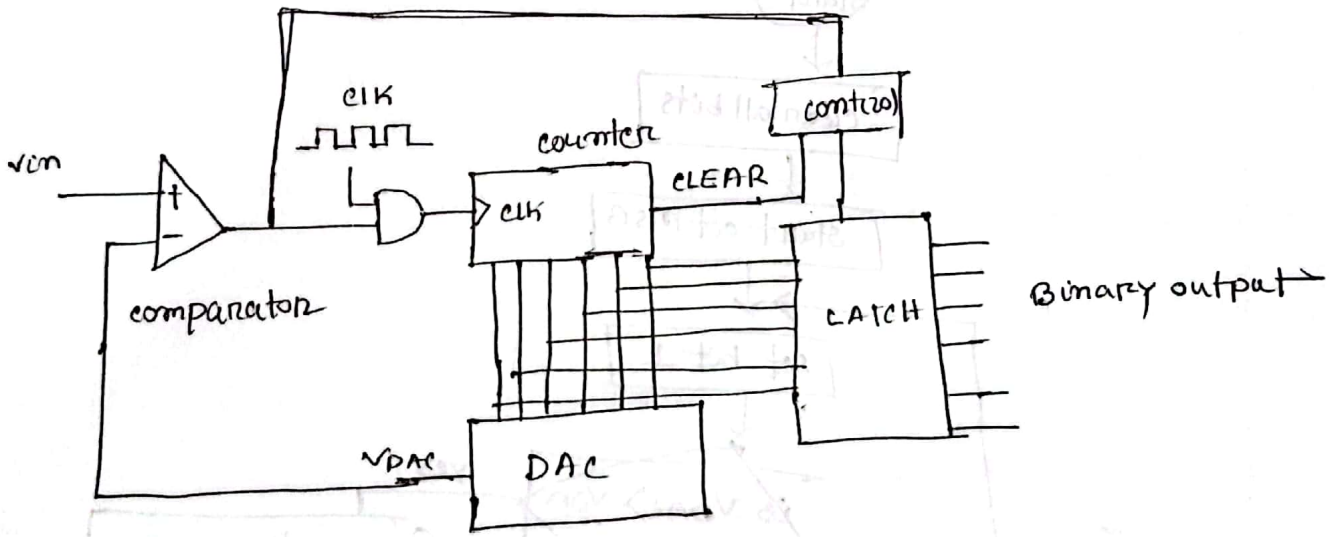
- For each iteration it will take 1 clock cycle. So for 4 bit ADC it will take 4 clock cycle.

conversion time $T_c = N \times T_{clk}$ (independent of the input voltage)

Advantages of SAR ADC

1. High Accuracy.
2. Low power consumption.
3. Easy to use.
4. Low latency time.

Digital Ramp ADC



$V_{in} > V_{DAC} \rightarrow$ high (1)

$V_{in} < V_{DAC} \rightarrow$ low (0)

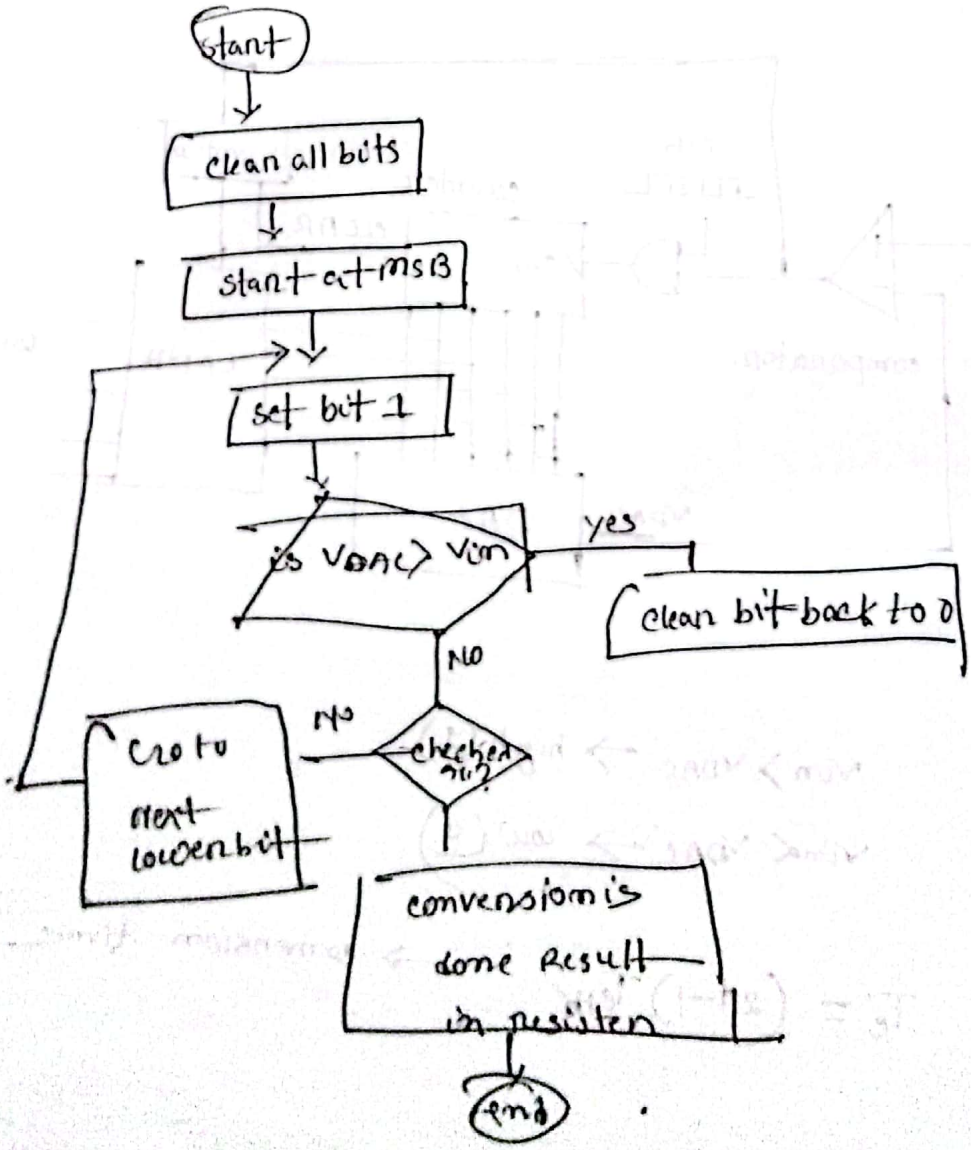
$T_c = (2^N - 1)\, T_{CLK}$ → conversion time.

(4) Resolution: Resolution of a D/A converter is defined as the smallest change that can occur in the analogue output as a result of change in the digital input.

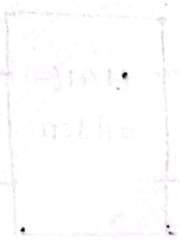
Number of different level $= 2^N$

" " steps $= 2^N - 1$..

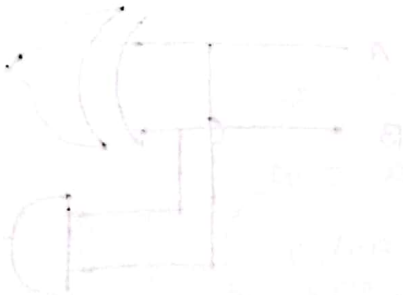
* % resolution $= \frac{\text{step size}}{\text{FSO}} \times 100\%$.



* Ramp ADC তে ক্ষেত্রে দশমিক আসলে পরের মান নিতে হবে আর SAC তে ক্ষেত্রে আগেরটি।

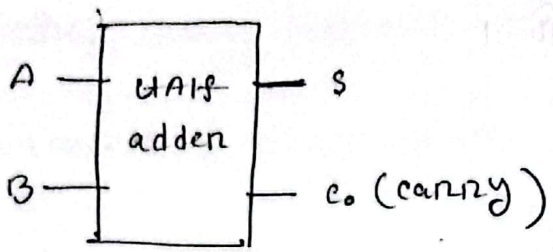






CamScanner

Half adder



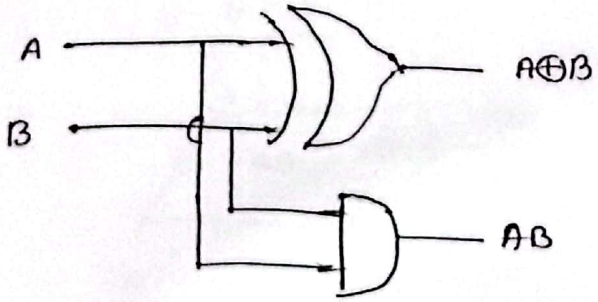
- It is used to add single bit numbers.
- It doesnot take carry from previous sum.

A	B	S	C_o
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

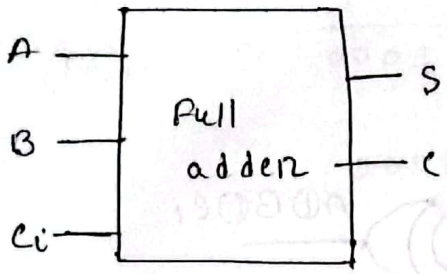
$S = \bar{A}B + A\bar{B}$

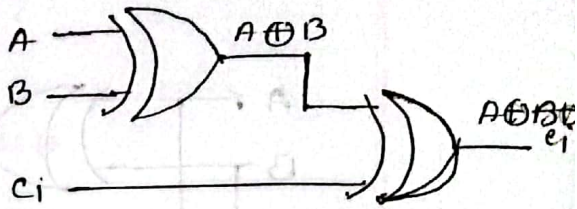
$= A \oplus B$

$C = AB$



Full adder





A	B	C_i	S	C
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

For carry

A \ BCi	00	01	11	10
0	0	0	1	0
1	0	1	1	1

$C_o = BC_i + AB + A\bar{C}i$

$C_o = AB + C_i(A \oplus B)$

$\Rightarrow AB + A\bar{B}C_i + \bar{A}BC_i$

A \ BCi	00	01	11	10
0	0	0	1	0
1	0	1	1	1

For sum

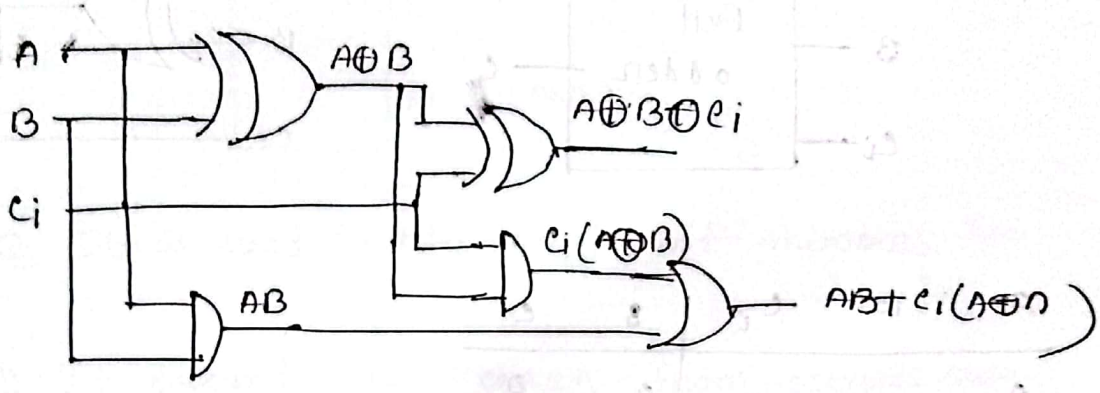
A \ BCi	00	01	11	10
0	0	1	0	1
1	1	0	1	0

check board conf

$S = A \oplus B \oplus C$

$S = A \oplus B + Ci$

$C = AB + Ci(A \oplus B)$



Full adder circuit

conversion

Binary coded decimal:

$(17)_{10} \rightarrow \underset{0001}{1} \quad \underset{0111}{7}$

$(00010111)_{BCD}$.

10100

$\underset{1}{0001} \quad \underset{4}{0100} = (14)_{10}$

Binary to grey code conversion

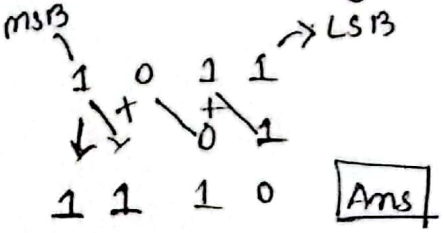
step-1: Record the MSB as it is.

step 2: Add the MSB to the next bit, record the sum and neglect the carry. (X-OR) → odd 1's detection — output 1.

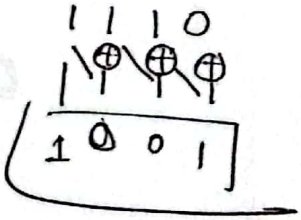
step:3: Repeat the process.

Ex-1: convert 1011 to grey code.

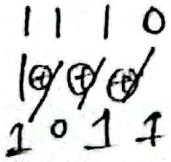
soln:



Ans



Grey to Bin:



Excess-3

Decimal — BCD 8-4-2-1 — Add 0011 — Excess (3).

$5 \longrightarrow 0101 \longrightarrow$ 1000 (8)

$$\begin{array}{r} 0101 \\ +\ 0011 \\ \hline 1000 \end{array}$$

$*(24)_{10}$ = 2 → 4

0010 0100
0011 0011
0101 0111 → (0101011) XS-3

~~0010 0100~~
0011
~~0101 0111~~

Excess-3

Excess-3

Excess-3

Pentonic strikes again

Pentonic is super-smooth.